PARITY-CONDITIONAL RANK POSSIBILITIES FOR ELLIPTIC CURVES Y^2=X^3-2PQX CLASSIFIED BY PRIME RESIDUE AND QUADRATIC SYMBOL

ARKABRATA GHOSH, BIDISHA ROY, RICHA SHARMA

Allowed standard prerequisites (editorial).

- 2-isogeny descent, local quartic solvability, quadratic reciprocity, root numbers and Parity Conjecture. Complete original formulations and bibliography retained; Parity Conjecture is an author-stated hypothesis, not supplied or certified.

1. Introduction

The study of elliptic curves plays an important role in modern number theory and arithmetic geometry. One of the most fundamental properties to an elliptic curve $E/\mathbb{Q}$ is the rank of its Mordell-Weil group $E(\mathbb{Q})$. Using the Mordell-Weil Theorem, we know the group of rational points of an elliptic curve is finitely generated and hence admits the following decomposition:

$$E(\mathbb{Q}) \cong E(\mathbb{Q})_{\text{tors}} \oplus \mathbb{Z}^r,$$

where $E(\mathbb{Q})_{\text{tors}}$ denotes the torsion subgroup and r is the Mordell–Weil rank. Even today, determining the rank of elliptic curves remains one of the most challenging problems in algebraic Number Theory and is closely related to the Birch and Swinnerton-Dyer conjecture.

A particularly fruitful approach to studying the rank of elliptic curves that possess a rational point of order two is the method of 2-descent. This method allows one to obtain information about the Mordell-Weil group through the analysis of certain associated homogeneous spaces (or torsors). Over the years, 2-descent has been successfully applied to several families of elliptic curves of the form $y^2 = x^3 - Dx$, leading to explicit rank computations and descriptions of their arithmetic properties. Spearman [12, 13] investigated the curves $y^2 = x^3 - px$ and $y^2 = x^3 - 2px$, and obtained conditions on the prime p under which these curves have rank at least two. Fujita and Terai [6] considered the family $y^2 = x^3 - p^k x$, where p is prime and $k = 1, 2, 3$, and established necessary and sufficient conditions for the rank to be equal to one or two. More recently, in a couple of articles [7, 8], Ghosh investigated the family $y^2 = x^3 - 5pqx$, and $y^2 = x^3 - pqx$ and determined several properties of its torsion subgroup and rank. The techniques developed there relied heavily on the method of 2-descent and the analysis of the associated homogeneous spaces.

Motivated by these results, in this paper, we consider the family

$$E_{-2pq} : \ y^2 = x^3 - 2pqx, \tag{1.1}$$

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where p and q are distinct odd primes. Our aim is to obtain explicit conditional possibilities and upper bounds for the rank of this family. Now, assuming the Parity Conjecture 2.2, we will prove the following Theorems.

Theorem 1.1. *Suppose p and q are distinct odd primes such that $(p,q) = (1,1) \pmod 8$. Then*

$$\operatorname{rank}(E_{-2pq}(\mathbb{Q})) = \begin{cases} 1 \text{ or } 3 \text{ or } 5, & \text{if } \left(\frac{p}{q}\right) = 1, \\ 1 \text{ or } 3, & \text{if } \left(\frac{p}{q}\right) = -1. \end{cases}$$

Theorem 1.2. *Suppose p and q are distinct odd primes such that $(p,q) \equiv (1,3)$, $(3,1)$, $(1,5)$, $(5,1)$, $(1,7)$ or $(7,1) \pmod 8$. Then we have*

$$\operatorname{rank}(E_{-2pq}(\mathbb{Q})) = \begin{cases} 1 \text{ or } 3, & \text{if } \left(\frac{p}{q}\right) = 1, \\ 1, & \text{if } \left(\frac{p}{q}\right) = -1. \end{cases}$$

Theorem 1.3. *Suppose p and q are distinct odd primes such that $(p,q) \equiv (3,3)$, $(3,5)$, $(5,3)$, $(3,7)$, $(7,3)$, $(5,7)$, $(7,5)$ or $(5,5) \pmod 8$. Then we have*

$$\operatorname{rank}(E_{-2pq}(\mathbb{Q})) = 1$$

for both $\left(\frac{p}{q}\right) = 1$ and $\left(\frac{p}{q}\right) = -1$.

Theorem 1.4. *(i) Suppose p and q are distinct odd primes such that $(p,q) = (7,7) \pmod 8$. Then*

$$\operatorname{rank}(E_{-2pq}(\mathbb{Q})) = 1 \text{ or } 3$$

for both $\left(\frac{p}{q}\right) = 1$ and $\left(\frac{p}{q}\right) = -1$.

The following table verifies the hypotheses of Theorems 1.1-1.4.

Rank 1 examples												Rank 3 examples					
p	q	Rank	p	q	Rank	p	q	Rank	p	q	Rank	p	q	Rank	p	q	Rank
3	5	1	3	7	1	3	11	1	3	97	3	7	23	3	13	17	3
5	7	1	5	11	1	5	13	1	17	43	3	17	59	3	17	73	3
7	11	1	7	13	1	7	17	1	17	89	3	19	73	3	23	47	3
11	13	1	11	17	1	11	19	1	31	71	3	41	61	3	41	73	3
13	19	1	13	23	1	13	29	1	47	79	3	53	97	3	61	73	3
17	19	1	17	23	1	17	29	1	73	97	3	79	97	3	89	97	3

TABLE 1. Examples of ranks 1 and 3 for $E_{-2pq} : y^2 = x^3 - 2pqx$, where $3 \le p < q \le 97$.

In the next theorem, we provide analogous results of Theorem 1.3 without assuming the parity conjecture.

Theorem 1.5. (i) *Let $(p,q) \equiv (3,3) \pmod 8$ such that $(pq - 32)$ is a perfect square. Then $\operatorname{rank}(E_{-2pq}(\mathbb{Q})) = 1$ for $\left(\frac{p}{q}\right) = 1$ and $\left(\frac{p}{q}\right) = -1$.*

(ii) *If $(p,q) \equiv (3,5) \pmod 8$, then $\operatorname{rank}(E_{-2pq}(\mathbb{Q})) = 1$ whenever*

$$\begin{aligned} \left(\tfrac{p}{q}\right) &= 1 \quad \text{and } p - 2q \text{ is a perfect square or,} \\ \left(\tfrac{p}{q}\right) &= -1 \quad \text{and } 2p - q \text{ is a perfect square.} \end{aligned}$$

holds.

(iii) *If* $(p,q) \equiv (5,3) \pmod 8$, *then* $\operatorname{rank}(E_{-2pq}(\mathbb{Q})) = 1$ *whenever*

$$\begin{array}{ll} \left(\frac{p}{q}\right) = 1 & \text{and } q-2p \text{ is a perfect square, or} \\ \left(\frac{p}{q}\right) = -1 & \text{and } 2q-p \text{ is a perfect square.} \end{array}$$

holds.

(iv) *If* $(p,q) \equiv (3,7) \pmod 8$, *then* $\operatorname{rank}(E_{-2pq}(\mathbb{Q})) = 1$ *whenever*

$$\begin{array}{ll} \left(\frac{p}{q}\right) = 1 & \text{and } 32p-q \text{ is a perfect square, or} \\ \left(\frac{p}{q}\right) = -1 & \text{and } q-2p \text{ is a perfect square.} \end{array}$$

holds.

(v) *If* $(p,q) \equiv (7,3) \pmod 8$, *then* $\operatorname{rank}(E_{-2pq}(\mathbb{Q})) = 1$ *whenever*

$$\begin{array}{ll} \left(\frac{p}{q}\right) = 1 & \text{and } p-2q \text{ is a perfect square, or} \\ \left(\frac{p}{q}\right) = -1 & \text{and } 32q-p \text{ is a perfect square.} \end{array}$$

holds.

(vi) *If* $(p,q) \equiv (5,7) \pmod 8$, *then* $\operatorname{rank}(E_{-2pq}(\mathbb{Q})) = 1$ *whenever*

$$\begin{array}{ll} \left(\frac{p}{q}\right) = 1 & \text{and } 32p-q \text{ is a perfect square, or} \\ \left(\frac{p}{q}\right) = -1 & \text{and } 2q-p \text{ is a perfect square.} \end{array}$$

holds.

(vii) *If* $(p,q) \equiv (7,5) \pmod 8$, *then* $\operatorname{rank}(E_{-2pq}(\mathbb{Q})) = 1$ *whenever*

$$\begin{array}{ll} \left(\frac{p}{q}\right) = 1 & \text{and } 2p-q \text{ is a perfect square, or} \\ \left(\frac{p}{q}\right) = -1 & \text{and } 32q-p \text{ is a perfect square.} \end{array}$$

holds.

(viii) *If* $(p,q) \equiv (5,5) \pmod 8$ *and* $2pq-1$ *is a perfect square, then*

$$\operatorname{rank}(E_{-2pq}(\mathbb{Q})) = 1$$

for both $\left(\frac{p}{q}\right) = 1$ *and* $\left(\frac{p}{q}\right) = -1$.

2. Preliminaries

To compute the rank of the curve given by (1.1), we will use the method of 2-descent. We will describe it briefly here. See [14, Chapter III, Sections 4–6] for more details. Suppose that $E : Y^2 = X^3 + aX^2 + bX$ is an elliptic curve over $\mathbb{Q}$ and $\overline{E} : Y^2 = X^3 - 2aX^2 + (a^2 - 4b)\,X$ is the corresponding isogenous curve to E. Hence, there exists an isogeny $\phi : E \to \overline{E}$ of degree 2 given by

$$\phi(x,y) = \left(\frac{y^2}{x^2}, \frac{y(x^2-b)}{x^2}\right).$$

Moreover, let $\mathbb{Q}^*$ be the multiplicative group of all non-zero rational numbers, and $\mathbb{Q}^{*2}$ be its subgroup of squares of elements of $\mathbb{Q}^*$. Hence, $\mathbb{Q}^*/\mathbb{Q}^{*2}$ is the multiplicative group of

all non-zero rational numbers modulo squares. Now we define the 2-descent homomorphism $\alpha : E(\mathbb{Q}) \to \mathbb{Q}^*/\mathbb{Q}^{*2}$ by

$$\alpha(P) = \begin{cases} 1 \pmod{\mathbb{Q}^{*2}}, \text{ if } P = \mathcal{O}, \text{ the point at infinity,} \\ b \pmod{\mathbb{Q}^{*2}}, \text{ if } P = (0,0), \\ x \pmod{\mathbb{Q}^{*2}}, \text{ if } P = (x,y) \text{ with } x \neq 0. \end{cases}$$

Similarly, we can define the 2-decent homomorphism on the isogenous curve $\overline{E}(\mathbb{Q})$ as follows: $\overline{\alpha} : \overline{E}(\mathbb{Q}) \to \mathbb{Q}^*/\mathbb{Q}^{*2}$ by

$$\overline{\alpha}(\overline{P}) = \begin{cases} 1 \pmod{\mathbb{Q}^{*2}}, \text{ if } \overline{P} = \overline{\mathcal{O}}, \text{ the point at infinity,} \\ \overline{b} \pmod{\mathbb{Q}^{*2}}, \text{ if } \overline{P} = (0,0), \\ x \pmod{\mathbb{Q}^{*2}}, \text{ if } \overline{P} = (x,y) \text{ with } x \neq 0, \end{cases}$$

where $\overline{b} = a^2 - 4b$.

The group $\alpha(E(\mathbb{Q}))$ consists of $1, b$ and all factors $b_1 \neq 1, b$ of b taken modulo $\mathbb{Q}^{*2}$ that satisfy the following conditions. For $b_1 \neq 1$, or $b \pmod{\mathbb{Q}^{*2}}$, there exists a triple $(N, M, e) \in \mathbb{Z}^3$, where $M \neq 0, e \neq 0$ is a solution of the Diophantine equation (or 'torsor')

$$\mathcal{T} : N^2 = b_1 M^4 + aM^2e^2 + b_2 e^4, \text{ with } b_1 b_2 = b,$$

and satisfies the following criterion:

$$\gcd(N, e) = \gcd(M, e) = \gcd(b_1, e) = \gcd(b_2, M) = \gcd(M, N) = 1. \tag{2.1}$$

Similarly, the group $\overline{\alpha}\left(\overline{E}(\mathbb{Q})\right)$ consists of $1, a^2 - 4b$ and all factors b_1 of $a^2 - 4b$ taken modulo $\mathbb{Q}^{*2}$ that satisfy the following conditions. For $b_1 \neq 1$, or $a^2 - 4b \pmod{\mathbb{Q}^{*2}}$, there exists a triple $(N, M, e) \in \mathbb{Z}^3$, where $M \neq 0, e \neq 0$ is a solution of the Diophantine equation (or 'torsor'),

$$\overline{\mathcal{T}} : N^2 = b_1 M^4 - 2aM^2e^2 + b_2 e^4, \text{ with } b_1 b_2 = a^2 - 4b,$$

and the same gcd criterion as in (2.1) above holds.

Now, to compute the rank of an elliptic curve, E, of the above form, we use the following proposition (see the displayed equation at the bottom of page 91 of [14]).

Proposition 2.1. *Let r be the rank of $E(\mathbb{Q})$, where E, α and $\overline{\alpha}$ are as above. Then,*

$$\frac{1}{4}|\alpha(E(\mathbb{Q}))| \cdot \left|\overline{\alpha}\left(\overline{E}(\mathbb{Q})\right)\right| = 2^r.$$

Next, we recall another concept called *root number* for an elliptic curve, E, defined over $\mathbb{Q}$. It is well-known that there is an L-function associated with it, $L(E/\mathbb{Q}, s)$. It satisfies the functional equation

$$L^*(E/\mathbb{Q}, 2 - s) = w(E/\mathbb{Q})L^*(E/\mathbb{Q}, s),$$

where $w(E/\mathbb{Q}) = \pm 1$ is called the *global root number*. The global root number $w(E/\mathbb{Q})$ is defined as the product of *local root numbers* $w\left(E/\mathbb{Q}_v\right) \in \{\pm 1\}$,

$$w(E/\mathbb{Q}) = \prod_v w\left(E/\mathbb{Q}_v\right),$$

where the product runs over all places v of $\mathbb{Q}$, including the infinite one. For more information about root numbers, see [4]. We list the following lemmas, which will be useful to prove the main results.

Lemma 2.1. *Let p and q be distinct odd primes. Then*

$$w\left(E_{-2pq}/\mathbb{Q}\right) = -1.$$

Proof. Put $D = 2pq$, so that E_{-2pq} is the curve $E_D : y^2 = x^3 - Dx$. By the formula of Birch and Stephens [2, (9) and (10), p. 296],

$$w(E_D/\mathbb{Q}) = w(E_D/\mathbb{R})w(E_D/\mathbb{Q}_2)\prod_{\ell^2 \| D} w(E_D/\mathbb{Q}_\ell).$$

Since $D > 0$, we have $w(E_D/\mathbb{R}) = \mathrm{sign}(-D) = -1$. Moreover, $D = 2pq$ is even, and hence it is not congruent to $1, 3, 11$, or $13 \pmod{16}$; therefore $w(E_D/\mathbb{Q}_2) = 1$. Finally, D is square-free, so there is no odd prime ℓ satisfying $\ell^2 \parallel D$, and the product is empty. Consequently, $w(E_{-2pq}/\mathbb{Q}) = -1$. □

Conjecture 2.2 (Parity Conjecture)**.** *Let $E/\mathbb{Q}$ be an elliptic curve defined over $\mathbb{Q}$. Then*

$$(-1)^{\mathrm{rank}(E(\mathbb{Q}))} = w(E/\mathbb{Q}).$$

Lemma 2.2. *([5], Theorem 1.2) Let $E/\mathbb{Q}$ be an elliptic curve defined over $\mathbb{Q}$. If the Tate-Shafarevich group, $Ш(E/\mathbb{Q})$, is finite, then the Parity Conjecture holds.*

Lemma 2.3. *Let $E/\mathbb{Q}$ be an elliptic curve defined over $\mathbb{Q}$. If $\mathrm{ord}_{s=1} L(E/\mathbb{Q}, s) \leq 1$, then $\mathrm{rank}(E(\mathbb{Q})) = \mathrm{ord}_{s=1} L(E/\mathbb{Q}, s)$ and $Ш(E/\mathbb{Q})$, is finite.*

3. Rank of E_{-2pq}

3.1. Throughout this section, we assume that $p \neq q$ are distinct odd primes. To compute the rank of E_{-2pq}, we first determine $\alpha(E_{-2pq}(\mathbb{Q}))$. By definition, $1, -2pq \in \alpha(E_{-2pq}(\mathbb{Q}))$.

Since $\alpha(E_{-2pq}(\mathbb{Q}))$ is a subgroup of $\mathbb{Q}^*/\mathbb{Q}^{*2}$ containing $-2pq$, the two square classes d and $-2pq/d$ occur simultaneously. Moreover, the torsors corresponding to the factorizations $-2pq = b_1 b_2$ and $-2pq = b_2 b_1$ are obtained from one another by interchanging M and e. Hence it is enough to choose one representative from each complementary pair. We take

$$T = \{-1, 2, -2, q, -q, p, -p\}.$$

The complementary classes are, respectively, $2pq, -pq, pq, -2p, 2p, -2q$, and $2q$.

We therefore need to consider only the following seven torsors over the integers.

b_1	b_2	Torsor
-1	$2pq$	$\mathcal{T}_1 :\ N^2 = -M^4 + 2pqe^4$
2	$-pq$	$\mathcal{T}_2 :\ N^2 = 2M^4 - pqe^4$
-2	pq	$\mathcal{T}_3 :\ N^2 = -2M^4 + pqe^4$
q	$-2p$	$\mathcal{T}_4 :\ N^2 = qM^4 - 2pe^4$
$-q$	$2p$	$\mathcal{T}_5 :\ N^2 = -qM^4 + 2pe^4$
p	$-2q$	$\mathcal{T}_6 :\ N^2 = pM^4 - 2qe^4$
$-p$	$2q$	$\mathcal{T}_7 :\ N^2 = -pM^4 + 2qe^4$

Table 2. Torsors for E_{-2pq}.

Lemma 3.1. *There exist integer solutions for the torsor* $\mathcal{T}_1 : N^2 = -M^4 + 2pqe^4$ *satisfying* (2.1) *only if* $p \equiv q \equiv 1 \pmod 4$ *and* $pq \equiv 1,\ 5 \pmod 8$.

Proof. Reducing $\mathcal{T}_1$ modulo p, we get $N^2 \equiv -M^4 \pmod p$. Since $p \nmid M$ by (2.1), we get $\left(\frac{-1}{p}\right) = 1$, and hence $p \equiv 1 \pmod 4$. Similarly, reducing $\mathcal{T}_1$ modulo q, we get $\left(\frac{-1}{q}\right) = 1$, and hence $q \equiv 1 \pmod 4$.

Now reducing $\mathcal{T}_1$ modulo 16, from (2.1), we deduce that M is odd and hence N is odd. Thus $N^2 \equiv 1,\ 9 \pmod{16}$ and $M^4 \equiv 1 \pmod{16}$. If e is even, then $e^4 \equiv 0 \pmod{16}$, and so $N^2 \equiv -1 \pmod{16}$, which is impossible. Hence e is odd. Therefore $e^4 \equiv 1 \pmod{16}$, and using this in $N^2 = -M^4 + 2pqe^4$, we get $2pq - 1 \equiv 1,\ 9 \pmod{16}$. After simplification, we obtain $pq \equiv 1,\ 5 \pmod 8$. □

Lemma 3.2. *There exist integer solutions for the torsor* $\mathcal{T}_2 : N^2 = 2M^4 - pqe^4$ *satisfying* (2.1) *only if* $\left(\frac{2}{p}\right) = 1$ *and* $\left(\frac{2}{q}\right) = 1$. *Equivalently,* $p, q \equiv 1$ *or* $7 \pmod 8$. *Moreover, reducing modulo* 16, *one obtains either* $pq \equiv 1,\ 9 \pmod{16}$ *or* $pq \equiv 7,\ 15 \pmod{16}$.

Proof. Reducing $\mathcal{T}_2$ modulo p, we get $N^2 \equiv 2M^4 \pmod p$. Since $p \nmid M$ by (2.1), we get $\left(\frac{2}{p}\right) = 1$, and hence $p \equiv 1$ or $7 \pmod 8$. Similarly, reducing $\mathcal{T}_2$ modulo q, we get $\left(\frac{2}{q}\right) = 1$, and hence $q \equiv 1$ or $7 \pmod 8$.

Now, from (2.1), we have e odd. Hence $e^4 \equiv 1 \pmod{16}$. If M is odd, then $M^4 \equiv 1 \pmod{16}$, and reducing $\mathcal{T}_2$ modulo 16, we get $2 - pq \equiv 1,\ 9 \pmod{16}$. Therefore $pq \equiv 1,\ 9 \pmod{16}$. If M is even, then $M^4 \equiv 0 \pmod{16}$, and so $-pq \equiv 1,\ 9 \pmod{16}$. Therefore $pq \equiv 7,\ 15 \pmod{16}$. □

Lemma 3.3. *There exist integer solutions for the torsor* $\mathcal{T}_3 : N^2 = -2M^4 + pqe^4$ *satisfying* (2.1) *only if* $\left(\frac{-2}{p}\right) = 1$ *and* $\left(\frac{-2}{q}\right) = 1$. *Equivalently,* $p, q \equiv 1$ *or* $3 \pmod 8$. *Moreover, reducing modulo* 16, *one obtains either* $pq \equiv 3,\ 11 \pmod{16}$ *or* $pq \equiv 1,\ 9 \pmod{16}$.

Proof. Reducing $\mathcal{T}_3$ modulo p, we get $N^2 \equiv -2M^4 \pmod p$. Since $p \nmid M$ by (2.1), we get $\left(\frac{-2}{p}\right) = 1$, and hence $p \equiv 1$ or $3 \pmod 8$. Similarly, reducing $\mathcal{T}_3$ modulo q, we get $\left(\frac{-2}{q}\right) = 1$, and hence $q \equiv 1$ or $3 \pmod 8$.

Now, from (2.1), we have e odd. Hence $e^4 \equiv 1 \pmod{16}$. If M is odd, then $M^4 \equiv 1 \pmod{16}$, and reducing $\mathcal{T}_3$ modulo 16, we get $-2+pq \equiv 1,\ 9 \pmod{16}$. Therefore $pq \equiv 3,\ 11 \pmod{16}$. If M is even, then $M^4 \equiv 0 \pmod{16}$, and so $pq \equiv 1,\ 9 \pmod{16}$. □

Lemma 3.4. *There exist integer solutions for the torsors* $\mathcal{T}_4 : N^2 = qM^4 - 2pe^4$ *and* $\mathcal{T}_5 : N^2 = -qM^4 + 2pe^4$ *satisfying* (2.1) *only under the following necessary conditions. For* $\mathcal{T}_4$, *one must have* $\left(\frac{q}{p}\right) = 1$ *and* $\left(\frac{-2p}{q}\right) = 1$, *with either* $q \equiv 1 \pmod 8$ *or* $q - 2p \equiv 1,\ 9$

(mod 16). *For* $\mathcal{T}_5$, *one must have* $\left(\frac{-q}{p}\right) = 1$ *and* $\left(\frac{2p}{q}\right) = 1$, *with either* $q \equiv 7 \pmod 8$ *or* $2p - q \equiv 1,\ 9 \pmod{16}$.

Proof. First consider $\mathcal{T}_4 : N^2 = qM^4 - 2pe^4$. Reducing $\mathcal{T}_4$ modulo p, we get $N^2 \equiv qM^4 \pmod p$. Since $p \nmid M$ by (2.1), we get $\left(\frac{q}{p}\right) = 1$. Reducing $\mathcal{T}_4$ modulo q, we get $N^2 \equiv -2pe^4 \pmod q$. Since $q \nmid e$ by (2.1), we get $\left(\frac{-2p}{q}\right) = 1$.

Now reducing $\mathcal{T}_4$ modulo 16, from (2.1), we deduce that M is odd and hence N is odd. Thus $N^2 \equiv 1,\ 9 \pmod{16}$ and $M^4 \equiv 1 \pmod{16}$. If e is even, then $e^4 \equiv 0 \pmod{16}$, and so $N^2 \equiv q \pmod{16}$. Therefore $q \equiv 1,\ 9 \pmod{16}$, which implies $q \equiv 1 \pmod 8$. If e is odd, then $e^4 \equiv 1 \pmod{16}$, and using this in $N^2 = qM^4 - 2pe^4$, we get $q - 2p \equiv 1,\ 9 \pmod{16}$.

Now consider $\mathcal{T}_5 : N^2 = -qM^4 + 2pe^4$. Reducing $\mathcal{T}_5$ modulo p, we get $N^2 \equiv -qM^4 \pmod p$. Since $p \nmid M$ by (2.1), we get $\left(\frac{-q}{p}\right) = 1$. Reducing $\mathcal{T}_5$ modulo q, we get $N^2 \equiv 2pe^4 \pmod q$. Since $q \nmid e$ by (2.1), we get $\left(\frac{2p}{q}\right) = 1$.

Again, reducing $\mathcal{T}_5$ modulo 16, from (2.1), we get that M is odd and hence N is odd. If e is even, then $e^4 \equiv 0 \pmod{16}$, and so $N^2 \equiv -q \pmod{16}$. Therefore $-q \equiv 1,\ 9 \pmod{16}$, which implies $q \equiv 7 \pmod 8$. If e is odd, then $e^4 \equiv 1 \pmod{16}$, and using this in $N^2 = -qM^4 + 2pe^4$, we get $2p - q \equiv 1,\ 9 \pmod{16}$. □

Lemma 3.5. *There exist integer solutions for the torsors* $\mathcal{T}_6 : N^2 = pM^4 - 2qe^4$ *and* $\mathcal{T}_7 : N^2 = -pM^4 + 2qe^4$ *satisfying* (2.1) *only under the following necessary conditions. For* $\mathcal{T}_6$, *one must have* $\left(\frac{p}{q}\right) = 1$ *and* $\left(\frac{-2q}{p}\right) = 1$, *with either* $p \equiv 1 \pmod 8$ *or* $p - 2q \equiv 1,\ 9 \pmod{16}$. *For* $\mathcal{T}_7$, *one must have* $\left(\frac{-p}{q}\right) = 1$ *and* $\left(\frac{2q}{p}\right) = 1$, *with either* $p \equiv 7 \pmod 8$ *or* $2q - p \equiv 1,\ 9 \pmod{16}$.

Proof. Interchanging p and q in Lemma 3.4 sends $\mathcal{T}_4$ to $\mathcal{T}_6$ and $\mathcal{T}_5$ to $\mathcal{T}_7$. Under the same interchange, the Legendre-symbol and congruence conditions in Lemma 3.4 become exactly the conditions stated above. Thus the result follows directly from Lemma 3.4.

□

Remark 3.1. *From Lemmas 3.1–3.5, we have seen that the possible extra elements of* $\alpha\left(E_{-2pq}(\mathbb{Q})\right)$ *can arise only from the torsors* $\mathcal{T}_1, \ldots, \mathcal{T}_7$. *Therefore, for each residue class of* (p, q) *modulo* 8, *we check the solvability conditions of these torsors and obtain an upper bound for* $\alpha\left(E_{-2pq}(\mathbb{Q})\right)$.

So, using Lemmas 3.1–3.5, we obtain the following results.

Case: I. At first, we shall consider the case when $\left(\frac{2}{p}\right) = 1 = \left(\frac{2}{q}\right)$. It gives $(p, q) \equiv (1, 1), (1, 7), (7, 1), (7, 7) \pmod 8$.

Lemma 3.6. *If* $(p,q) \equiv (1,1) \pmod 8$, *then*

$$\alpha\left(E_{-2pq}(\mathbb{Q})\right) \subseteq \begin{cases} \{1,-1,2,-2,p,-p,q,-q,2p,-2p,2q,-2q,pq,-pq,2pq,-2pq\}, & \text{for } \left(\frac{p}{q}\right)=1, \\ \{1,-1,2,-2,pq,-pq,2pq,-2pq\}, & \text{for } \left(\frac{p}{q}\right)=-1. \end{cases}$$

Proof. $\mathcal{T}_1$: Since $p \equiv q \equiv 1 \pmod 8$, Lemma 3.1 shows that $\mathcal{T}_1$ may have integer solutions. Hence -1 and $2pq$ may be elements of $\alpha\left(E_{-2pq}(\mathbb{Q})\right)$.

$\mathcal{T}_2$ and $\mathcal{T}_3$: Since $p \equiv q \equiv 1 \pmod 8$, the congruence conditions in Lemmas 3.2 and 3.3 are satisfied. Hence $2, -pq, -2, pq$ may be elements of $\alpha\left(E_{-2pq}(\mathbb{Q})\right)$.

$\mathcal{T}_4$ and $\mathcal{T}_5$: Here $\left(\frac{q}{p}\right) = \left(\frac{p}{q}\right)$ by quadratic reciprocity. If $\left(\frac{p}{q}\right) = 1$, then all the Legendre-symbol conditions in Lemma 3.4 are satisfied, and hence $q, -2p, -q, 2p$ may be elements of $\alpha\left(E_{-2pq}(\mathbb{Q})\right)$. If $\left(\frac{p}{q}\right) = -1$, then the Legendre-symbol conditions fail, and hence these torsors have no integer solutions.

$\mathcal{T}_6$ and $\mathcal{T}_7$: Similarly, if $\left(\frac{p}{q}\right) = 1$, then all the Legendre-symbol conditions in Lemma 3.5 are satisfied, and hence $p, -2q, -p, 2q$ may be elements of $\alpha\left(E_{-2pq}(\mathbb{Q})\right)$. If $\left(\frac{p}{q}\right) = -1$, then these torsors have no integer solutions.

Combining the above possibilities, we get the desired result. □

Lemma 3.7. *If* $(p,q) \equiv (1,7) \pmod 8$, *then*

$$\alpha\left(E_{-2pq}(\mathbb{Q})\right) \subseteq \begin{cases} \{1,2,p,-q,2p,-2q,-pq,-2pq\}, & \text{for } \left(\frac{p}{q}\right)=1, \\ \{1,2,-pq,-2pq\}, & \text{for } \left(\frac{p}{q}\right)=-1. \end{cases}$$

Lemma 3.8. *If* $(p,q) \equiv (7,1) \pmod 8$, *then*

$$\alpha\left(E_{-2pq}(\mathbb{Q})\right) \subseteq \begin{cases} \{1,2,-p,q,-2p,2q,-pq,-2pq\}, & \text{for } \left(\frac{p}{q}\right)=1, \\ \{1,2,-pq,-2pq\}, & \text{for } \left(\frac{p}{q}\right)=-1. \end{cases}$$

Proof. This is obtained from Lemma 3.7 by interchanging p and q. □

Lemma 3.9. *If* $(p,q) \equiv (7,7) \pmod 8$, *then*

$$\alpha\left(E_{-2pq}(\mathbb{Q})\right) \subseteq \begin{cases} \{1,2,p,-q,2p,-2q,-pq,-2pq\}, & \text{for } \left(\frac{p}{q}\right)=1, \\ \{1,2,-p,q,-2p,2q,-pq,-2pq\}, & \text{for } \left(\frac{p}{q}\right)=-1. \end{cases}$$

Case: II. Now we shall consider the case when $\left(\frac{2}{p}\right) = 1$ and $\left(\frac{2}{q}\right) = -1$. It gives $(p,q) \equiv (1,3), (1,5), (7,3), (7,5) \pmod 8$.

Lemma 3.10. *If* $(p,q) \equiv (1,3) \pmod 8$, *then*

$$\alpha\left(E_{-2pq}(\mathbb{Q})\right) \subseteq \begin{cases} \{1,-2,p,q,-2p,-2q,pq,-2pq\}, & \text{for } \left(\frac{p}{q}\right)=1, \\ \{1,-2,pq,-2pq\}, & \text{for } \left(\frac{p}{q}\right)=-1. \end{cases}$$

Proof. $\mathcal{T}_1$: Since $q \equiv 3 \pmod 8$, Lemma 3.1 shows that $\mathcal{T}_1$ has no integer solutions.

$\mathcal{T}_2$: Since $q \equiv 3 \pmod 8$, Lemma 3.2 shows that $\mathcal{T}_2$ has no integer solutions.

$\mathcal{T}_3$: Since $p \equiv 1 \pmod 8$ and $q \equiv 3 \pmod 8$, Lemma 3.3 shows that $\mathcal{T}_3$ may have integer solutions. Hence -2 and pq may be elements of $\alpha\left(E_{-2pq}(\mathbb{Q})\right)$.

$\mathcal{T}_4$: Since $p \equiv 1 \pmod 8$, by quadratic reciprocity we have $\left(\frac{q}{p}\right) = \left(\frac{p}{q}\right)$. Moreover, since $q \equiv 3 \pmod 8$, we have $\left(\frac{-2}{q}\right) = 1$. Hence $\left(\frac{-2p}{q}\right) = \left(\frac{p}{q}\right)$. Therefore, if $\left(\frac{p}{q}\right) = 1$, then the Legendre-symbol conditions of $\mathcal{T}_4$ are satisfied. Also $q - 2p \equiv 1 \pmod 8$, so the congruence condition in Lemma 3.4 may hold. Hence q and $-2p$ may be elements of $\alpha\left(E_{-2pq}(\mathbb{Q})\right)$. If $\left(\frac{p}{q}\right) = -1$, then the Legendre-symbol conditions fail, so $\mathcal{T}_4$ has no integer solutions.

$\mathcal{T}_5$: Since $q \equiv 3 \pmod 8$, the congruence condition in Lemma 3.4 for $\mathcal{T}_5$ is not satisfied. Hence $\mathcal{T}_5$ has no integer solutions.

$\mathcal{T}_6$: If $\left(\frac{p}{q}\right) = 1$, then the Legendre-symbol conditions in Lemma 3.5 are satisfied, and since $p \equiv 1 \pmod 8$, the congruence condition also holds. Hence p and $-2q$ may be elements of $\alpha\left(E_{-2pq}(\mathbb{Q})\right)$. If $\left(\frac{p}{q}\right) = -1$, then $\mathcal{T}_6$ has no integer solutions.

$\mathcal{T}_7$: Since $p \equiv 1 \pmod 8$, the congruence condition in Lemma 3.5 for $\mathcal{T}_7$ is not satisfied. Hence $\mathcal{T}_7$ has no integer solutions.

Combining the above possibilities, we get the desired result. □

Lemma 3.11. *If* $(p, q) \equiv (1, 5) \pmod 8$, *then*

$$\alpha\left(E_{-2pq}(\mathbb{Q})\right) \subseteq \begin{cases} \{1, -1, p, -p, 2q, -2q, 2pq, -2pq\}, & \text{for } \left(\frac{p}{q}\right) = 1, \\ \{1, -1, 2pq, -2pq\}, & \text{for } \left(\frac{p}{q}\right) = -1. \end{cases}$$

Lemma 3.12. *If* $(p, q) \equiv (7, 3) \pmod 8$, *then*

$$\alpha\left(E_{-2pq}(\mathbb{Q})\right) \subseteq \begin{cases} \{1, p, -2q, -2pq\}, & \text{for } \left(\frac{p}{q}\right) = 1, \\ \{1, -p, 2q, -2pq\}, & \text{for } \left(\frac{p}{q}\right) = -1. \end{cases}$$

Lemma 3.13. *If* $(p, q) \equiv (7, 5) \pmod 8$, *then*

$$\alpha\left(E_{-2pq}(\mathbb{Q})\right) \subseteq \begin{cases} \{1, -p, 2q, -2pq\}, & \text{for } \left(\frac{p}{q}\right) = 1, \\ \{1, -q, 2p, -2pq\}, & \text{for } \left(\frac{p}{q}\right) = -1. \end{cases}$$

Case: III. Now we consider the case when $\left(\frac{2}{p}\right) = -1$ and $\left(\frac{2}{q}\right) = 1$, so that $(p, q) \equiv (3, 1), (3, 7), (5, 1), (5, 7) \pmod 8$. The four statements below are obtained, respectively, from Lemmas 3.10, 3.12, 3.11, and 3.13 by interchanging p and q. For the class $(3, 7)$, quadratic reciprocity reverses the sign of the Legendre symbol; in the other three classes, it preserves the sign.

Lemma 3.14. *If* $(p, q) \equiv (3, 1) \pmod 8$, *then*

$$\alpha\left(E_{-2pq}(\mathbb{Q})\right) \subseteq \begin{cases} \{1, -2, p, q, -2p, -2q, pq, -2pq\}, & \text{for } \left(\frac{p}{q}\right) = 1, \\ \{1, -2, pq, -2pq\}, & \text{for } \left(\frac{p}{q}\right) = -1. \end{cases}$$

Lemma 3.15. *If* $(p,q) \equiv (3,7) \pmod 8$, *then*

$$\alpha\left(E_{-2pq}(\mathbb{Q})\right) \subseteq \begin{cases} \{1,-q,2p,-2pq\}, & \text{for } \left(\frac{p}{q}\right)=1, \\ \{1,q,-2p,-2pq\}, & \text{for } \left(\frac{p}{q}\right)=-1. \end{cases}$$

Lemma 3.16. *If* $(p,q) \equiv (5,1) \pmod 8$, *then*

$$\alpha\left(E_{-2pq}(\mathbb{Q})\right) \subseteq \begin{cases} \{1,-1,q,-q,2p,-2p,2pq,-2pq\}, & \text{for } \left(\frac{p}{q}\right)=1, \\ \{1,-1,2pq,-2pq\}, & \text{for } \left(\frac{p}{q}\right)=-1. \end{cases}$$

Lemma 3.17. *If* $(p,q) \equiv (5,7) \pmod 8$, *then*

$$\alpha\left(E_{-2pq}(\mathbb{Q})\right) \subseteq \begin{cases} \{1,-q,2p,-2pq\}, & \text{for } \left(\frac{p}{q}\right)=1, \\ \{1,-p,2q,-2pq\}, & \text{for } \left(\frac{p}{q}\right)=-1. \end{cases}$$

Case: IV. Finally, we shall consider the case when $\left(\frac{2}{p}\right) = -1 = \left(\frac{2}{q}\right)$. It gives $(p,q) \equiv (3,3),(3,5),(5,3),(5,5) \pmod 8$.

Lemma 3.18. *If* $(p,q) \equiv (3,3) \pmod 8$, *then*

$$\alpha\left(E_{-2pq}(\mathbb{Q})\right) \subseteq \{1,-2,pq,-2pq\}$$

for both $\left(\frac{p}{q}\right)=1$ *and* $\left(\frac{p}{q}\right)=-1$.

Proof. Here $\mathcal{T}_1$ and $\mathcal{T}_2$ have no integer solutions by Lemmas 3.1 and 3.2. Since $p \equiv q \equiv 3 \pmod 8$, Lemma 3.3 shows that $\mathcal{T}_3$ may have integer solutions. Hence -2 and pq may be elements of $\alpha\left(E_{-2pq}(\mathbb{Q})\right)$. Also, Lemmas 3.4 and 3.5 show that $\mathcal{T}_4, \mathcal{T}_5, \mathcal{T}_6$ and $\mathcal{T}_7$ have no integer solutions. Therefore, we get the desired result. □

Lemma 3.19. *If* $(p,q) \equiv (3,5) \pmod 8$, *then*

$$\alpha\left(E_{-2pq}(\mathbb{Q})\right) \subseteq \begin{cases} \{1,p,-2q,-2pq\}, & \text{for } \left(\frac{p}{q}\right)=1, \\ \{1,-q,2p,-2pq\}, & \text{for } \left(\frac{p}{q}\right)=-1. \end{cases}$$

Proof. From Lemmas 3.1, 3.2 and 3.3, the torsors $\mathcal{T}_1, \mathcal{T}_2$ and $\mathcal{T}_3$ have no integer solutions. Also, $\mathcal{T}_4$ and $\mathcal{T}_7$ have no integer solutions. If $\left(\frac{p}{q}\right) = 1$, then $\mathcal{T}_6$ may have integer solutions, and hence p and $-2q$ may be elements of $\alpha\left(E_{-2pq}(\mathbb{Q})\right)$. If $\left(\frac{p}{q}\right) = -1$, then $\mathcal{T}_5$ may have integer solutions, and hence $-q$ and $2p$ may be elements of $\alpha\left(E_{-2pq}(\mathbb{Q})\right)$. Combining these possibilities, we get the desired result. □

Interchanging p and q, we get the following Lemma.

Lemma 3.20. *If* $(p,q) \equiv (5,3) \pmod 8$, *then*

$$\alpha\left(E_{-2pq}(\mathbb{Q})\right) \subseteq \begin{cases} \{1,q,-2p,-2pq\}, & \text{for } \left(\frac{p}{q}\right)=1, \\ \{1,-p,2q,-2pq\}, & \text{for } \left(\frac{p}{q}\right)=-1. \end{cases}$$

Lemma 3.21. *If* $(p,q) \equiv (5,5) \pmod 8$, *then*

$$\alpha\left(E_{-2pq}(\mathbb{Q})\right) \subseteq \{1,-1,2pq,-2pq\}$$

for both $\left(\frac{p}{q}\right) = 1$ *and* $\left(\frac{p}{q}\right) = -1$.

Proof. Since $p \equiv q \equiv 5 \pmod 8$, Lemma 3.1 shows that $\mathcal{T}_1$ may have integer solutions. Hence -1 and $2pq$ may be elements of $\alpha\left(E_{-2pq}(\mathbb{Q})\right)$. The remaining torsors $\mathcal{T}_2, \ldots, \mathcal{T}_7$ have no integer solutions by Lemmas 3.2, 3.3, 3.4 and 3.5. Therefore, we get the desired result. □

3.2. Now we determine $\left|\overline{\alpha}\left(\overline{E_{-2pq}}(\mathbb{Q})\right)\right|$. As $\overline{E_{-2pq}} : y^2 = x^3 + 8pqx$, we know that $1, 2pq \in \overline{\alpha}\left(\overline{E_{-2pq}}(\mathbb{Q})\right)$. Let us note the set of all possible divisors, b_1, of $8pq$ with $b_1 \neq 1, 2pq$ modulo $\mathbb{Q}^{*2}$ is

$$\overline{T} = \{2, p, q, 2p, 2q, pq\}.$$

We exclude all negative values of b_1 from the set $\overline{T}$ as the corresponding torsors $N^2 = b_1M^4 + b_2e^4$ will have no solutions if both b_1 and b_2 are negative.

Since $\overline{\alpha}\left(\overline{E_{-2pq}}(\mathbb{Q})\right)$ is a subgroup of $\mathbb{Q}^*/\mathbb{Q}^{*2}$ containing $2pq$, multiplication by $2pq$ gives the complementary pairs

$$\{2, pq\}, \qquad \{p, 2q\}, \qquad \{q, 2p\}.$$

Thus, it is enough to study one representative from each pair. We choose the representatives p, q, and pq, corresponding respectively to $\overline{\mathcal{T}_2}$, $\overline{\mathcal{T}_3}$, and $\overline{\mathcal{T}_6}$.

Label	b_1	b_2	Equation
$\overline{\mathcal{T}_1}$	2	$4pq$	$N^2 = 2M^4 + 4pqe^4$
$\overline{\mathcal{T}_2}$	p	$8q$	$N^2 = pM^4 + 8qe^4$
$\overline{\mathcal{T}_3}$	q	$8p$	$N^2 = qM^4 + 8pe^4$
$\overline{\mathcal{T}_4}$	$2p$	$4q$	$N^2 = 2pM^4 + 4qe^4$
$\overline{\mathcal{T}_5}$	$2q$	$4p$	$N^2 = 2qM^4 + 4pe^4$
$\overline{\mathcal{T}_6}$	pq	8	$N^2 = pqM^4 + 8e^4$

TABLE 3. Torsors for $\overline{E_{-2pq}}$.

Lemma 3.22. *The classes represented by* $\overline{\mathcal{T}_1}$ *and* $\overline{\mathcal{T}_6}$ *occur simultaneously in* $\overline{\alpha}\left(\overline{E_{-2pq}}(\mathbb{Q})\right)$. *Consequently, it is enough to study* $\overline{\mathcal{T}_6}$.

Proof. Since $2pq$ belongs to $\overline{\alpha}\left(\overline{E_{-2pq}}(\mathbb{Q})\right)$, we have

$$2 \in \overline{\alpha}\left(\overline{E_{-2pq}}(\mathbb{Q})\right) \iff 2(2pq) \equiv pq \pmod{\mathbb{Q}^{*2}} \in \overline{\alpha}\left(\overline{E_{-2pq}}(\mathbb{Q})\right).$$

The classes 2 and pq correspond to $\overline{\mathcal{T}_1}$ and $\overline{\mathcal{T}_6}$, respectively. □

Lemma 3.23. *There exist integer solutions for the torsor* $\overline{\mathcal{T}_2} : N^2 = pM^4 + 8qe^4$ *only if* $\left(\frac{p}{q}\right) = 1$, $\left(\frac{2q}{p}\right) = 1$, *and* $p \equiv 1 \pmod 8$.

Proof. Reducing $\overline{\mathcal{T}_2}$ modulo q and p, respectively, gives $\left(\frac{p}{q}\right) = 1$ and $\left(\frac{2q}{p}\right) = 1$. By (2.1), M is odd. Hence N is also odd, and reducing the torsor modulo 8 gives

$$1 \equiv N^2 \equiv pM^4 \equiv p \pmod 8.$$

□

Lemma 3.24. *There exist integer solutions for the torsor* $\overline{\mathcal{T}_3} : N^2 = qM^4 + 8pe^4$ *only if* $\left(\frac{q}{p}\right) = 1$, $\left(\frac{2p}{q}\right) = 1$, *and* $q \equiv 1 \pmod 8$.

Proof. This follows from Lemma 3.23 by interchanging p and q. □

Lemma 3.25. *The classes represented by* $\overline{\mathcal{T}_4}$ *and* $\overline{\mathcal{T}_3}$ *occur simultaneously, and the classes represented by* $\overline{\mathcal{T}_5}$ *and* $\overline{\mathcal{T}_2}$ *occur simultaneously, in* $\overline{\alpha}\left(\overline{E_{-2pq}}(\mathbb{Q})\right)$. *Consequently, no separate analysis of* $\overline{\mathcal{T}_4}$ *and* $\overline{\mathcal{T}_5}$ *is required.*

Proof. Multiplication by $2pq$ gives

$$(2p)(2pq) \equiv q \qquad \text{and} \qquad (2q)(2pq) \equiv p \pmod{\mathbb{Q}^{*2}}.$$

Thus the pair $\{2p, q\}$ corresponds to $\{\overline{\mathcal{T}_4}, \overline{\mathcal{T}_3}\}$, while the pair $\{2q, p\}$ corresponds to $\{\overline{\mathcal{T}_5}, \overline{\mathcal{T}_2}\}$. □

Lemma 3.26. *There exist integer solutions for the torsor* $\overline{\mathcal{T}_6} : N^2 = pqM^4 + 8e^4$ *only if* $\left(\frac{2}{p}\right) = 1$, $\left(\frac{2}{q}\right) = 1$, *and* $pq \equiv 1 \pmod 8$.

Proof. Reducing $\overline{\mathcal{T}_6}$ modulo p and q, respectively, gives $\left(\frac{2}{p}\right) = 1$ and $\left(\frac{2}{q}\right) = 1$. By (2.1), M is odd, and hence N is odd. Reducing the torsor modulo 8, we obtain

$$1 \equiv N^2 \equiv pqM^4 \equiv pq \pmod 8.$$

□

Remark 3.2. *By Lemmas 3.22 and 3.25, the six possible classes occur in three complementary pairs. So, for elements of* $\overline{\alpha}\left(\overline{E_{-2pq}}(\mathbb{Q})\right)$, *it is enough to check the solution of the torsor* $\overline{\mathcal{T}_2}$, $\overline{\mathcal{T}_3}$ *and* $\overline{\mathcal{T}_6}$.

So, using Lemmas 3.23, 3.24, and 3.26, we obtain the following results.

Case-I:. At first, we shall consider the case when $\left(\frac{2}{p}\right) = 1 = \left(\frac{2}{q}\right) = 1$. It gives $(p, q) \equiv (1, 1), (1, 7), (7, 1), (7, 7) \pmod 8$.

Lemma 3.27. *If* $(p, q) \equiv (1, 1) \pmod 8$, *then*

$$\overline{\alpha}\left(\overline{E_{-2pq}}(\mathbb{Q})\right) \subseteq \begin{cases} \{1, 2, p, q, 2p, 2q, pq, 2pq\}, & \text{for } \left(\frac{p}{q}\right) = 1, \\ \{1, 2, pq, 2pq\}, & \text{for } \left(\frac{p}{q}\right) = -1. \end{cases}$$

Proof. Since $p \equiv q \equiv 1 \pmod 8$, the conditions in Lemmas 3.23 and 3.24 are equivalent to $\left(\frac{p}{q}\right) = 1$, by quadratic reciprocity. Moreover, all the conditions in Lemma 3.26 are satisfied. Thus, for $\left(\frac{p}{q}\right) = 1$, the three complementary pairs may occur, whereas for $\left(\frac{p}{q}\right) = -1$, only the pair $\{2, pq\}$ may occur. This gives the stated inclusions. □

Lemma 3.28. *If* $(p,q) \equiv (1,7) \pmod 8$, *then*

$$\overline{\alpha}\left(\overline{E_{-2pq}}(\mathbb{Q})\right) \begin{cases} \subseteq \{1,p,2q,2pq\}, & \text{for } \left(\frac{p}{q}\right) = 1, \\ = \{1,2pq\}, & \text{for } \left(\frac{p}{q}\right) = -1. \end{cases}$$

Proof. Here Lemma 3.23 may contribute the pair $\{p,2q\}$ precisely when $\left(\frac{p}{q}\right) = 1$. Lemma 3.24 does not apply because $q \not\equiv 1 \pmod 8$, and Lemma 3.26 does not apply because $pq \not\equiv 1 \pmod 8$. Hence the result follows. □

Lemma 3.29. *If* $(p,q) \equiv (7,1) \pmod 8$, *then*

$$\overline{\alpha}\left(\overline{E_{-2pq}}(\mathbb{Q})\right) \begin{cases} \subseteq \{1,q,2p,2pq\}, & \text{for } \left(\frac{p}{q}\right) = 1, \\ = \{1,2pq\}, & \text{for } \left(\frac{p}{q}\right) = -1. \end{cases}$$

Proof. This follows from Lemma 3.28 by interchanging p and q. □

Lemma 3.30. *If* $(p,q) \equiv (7,7) \pmod 8$, *then*

$$\overline{\alpha}\left(\overline{E_{-2pq}}(\mathbb{Q})\right) \subseteq \{1,2,pq,2pq\}$$

for both $\left(\frac{p}{q}\right) = 1$ *and* $\left(\frac{p}{q}\right) = -1$.

Proof. Lemmas 3.23 and 3.24 do not apply because $p \not\equiv 1 \pmod 8$ and $q \not\equiv 1 \pmod 8$, respectively. On the other hand, all the conditions in Lemma 3.26 are satisfied. Hence only the pair $\{2,pq\}$ may occur, independently of $\left(\frac{p}{q}\right)$. □

Case-II:. Now we shall consider the case when $\left(\frac{2}{p}\right) = 1$ and $\left(\frac{2}{q}\right) = -1$. It gives $(p,q) \equiv (1,3),(1,5),(7,3),(7,5) \pmod 8$.

Lemma 3.31. *If* $(p,q) \equiv (1,3) \pmod 8$, *then*

$$\overline{\alpha}\left(\overline{E_{-2pq}}(\mathbb{Q})\right) \begin{cases} \subseteq \{1,p,2q,2pq\}, & \text{for } \left(\frac{p}{q}\right) = 1, \\ = \{1,2pq\}, & \text{for } \left(\frac{p}{q}\right) = -1. \end{cases}$$

Proof. Since $p \equiv 1 \pmod 8$, Lemma 3.23 may contribute the pair $\{p,2q\}$, and its Legendre-symbol conditions are equivalent to $\left(\frac{p}{q}\right) = 1$. Lemma 3.24 does not apply because $q \not\equiv 1 \pmod 8$, while Lemma 3.26 does not apply because $\left(\frac{2}{q}\right) = -1$. Hence the result follows. □

Lemma 3.32. *If* $(p,q) \equiv (1,5) \pmod 8$, *then*

$$\overline{\alpha}\left(\overline{E_{-2pq}}(\mathbb{Q})\right) \begin{cases} \subseteq \{1,p,2q,2pq\}, & \text{for } \left(\frac{p}{q}\right) = 1, \\ = \{1,2pq\}, & \text{for } \left(\frac{p}{q}\right) = -1. \end{cases}$$

Proof. The proof is identical to that of Lemma 3.31. □

Lemma 3.33. *If* $(p,q) \equiv (7,3) \pmod 8$, *then*

$$\overline{\alpha}\left(\overline{E_{-2pq}}(\mathbb{Q})\right) = \{1, 2pq\} \quad \text{for both } \left(\frac{p}{q}\right) = 1 \quad \text{and } \left(\frac{p}{q}\right) = -1.$$

Proof. Lemma 3.23 does not apply because $p \not\equiv 1 \pmod 8$, Lemma 3.24 does not apply because $q \not\equiv 1 \pmod 8$, and Lemma 3.26 does not apply because $\left(\frac{2}{q}\right) = -1$. Thus no additional complementary pair can occur. □

Lemma 3.34. *If* $(p,q) \equiv (7,5) \pmod 8$, *then*

$$\overline{\alpha}\left(\overline{E_{-2pq}}(\mathbb{Q})\right) = \{1, 2pq\} \quad \text{for both } \left(\frac{p}{q}\right) = 1 \quad \text{and } \left(\frac{p}{q}\right) = -1.$$

Proof. The proof is identical to that of Lemma 3.33. □

Case-III:. Now we consider the case when $\left(\frac{2}{p}\right) = -1$ and $\left(\frac{2}{q}\right) = 1$. It gives $(p,q) \equiv (3,1), (3,7), (5,1), (5,7) \pmod 8$.

These four cases are obtained from Case-II by interchanging p and q that is following Lemmas 3.31-3.34.

Lemma 3.35. *If* $(p,q) \equiv (3,1) \pmod 8$, *then*

$$\overline{\alpha}\left(\overline{E_{-2pq}}(\mathbb{Q})\right) \begin{cases} \subseteq \{1, q, 2p, 2pq\}, & \text{for } \left(\frac{p}{q}\right) = 1, \\ = \{1, 2pq\}, & \text{for } \left(\frac{p}{q}\right) = -1. \end{cases}$$

Lemma 3.36. *If* $(p,q) \equiv (3,7) \pmod 8$, *then*

$$\overline{\alpha}\left(\overline{E_{-2pq}}(\mathbb{Q})\right) = \{1, 2pq\} \quad \text{for both } \left(\frac{p}{q}\right) = 1 \quad \text{and } \left(\frac{p}{q}\right) = -1.$$

Lemma 3.37. *If* $(p,q) \equiv (5,1) \pmod 8$, *then*

$$\overline{\alpha}\left(\overline{E_{-2pq}}(\mathbb{Q})\right) \begin{cases} \subseteq \{1, q, 2p, 2pq\}, & \text{for } \left(\frac{p}{q}\right) = 1, \\ = \{1, 2pq\}, & \text{for } \left(\frac{p}{q}\right) = -1. \end{cases}$$

Lemma 3.38. *If* $(p,q) \equiv (5,7) \pmod 8$, *then*

$$\overline{\alpha}\left(\overline{E_{-2pq}}(\mathbb{Q})\right) = \{1, 2pq\} \quad \text{for both } \left(\frac{p}{q}\right) = 1 \quad \text{and } \left(\frac{p}{q}\right) = -1.$$

Case IV:. Finally, we shall consider the case when $\left(\frac{2}{p}\right) = -1$ and $\left(\frac{2}{q}\right) = -1$. It gives $(p,q) \equiv (3,3), (3,5), (5,3), (5,5) \pmod 8$.

In all four residue classes, neither p nor q is congruent to $1 \pmod 8$; therefore Lemmas 3.23 and 3.24 do not apply. Moreover, $\left(\frac{2}{p}\right) = \left(\frac{2}{q}\right) = -1$, so Lemma 3.26 does not apply. Thus the same argument proves all four lemmas below.

Lemma 3.39. *If* $(p,q) \equiv (3,3) \pmod 8$, *then*

$$\overline{\alpha}\left(\overline{E_{-2pq}}(\mathbb{Q})\right) = \{1, 2pq\} \quad \textit{for both } \left(\frac{p}{q}\right) = 1 \quad \textit{and } \left(\frac{p}{q}\right) = -1.$$

Proof. The preceding observation rules out all three representative torsors $\overline{\mathcal{T}_2}$, $\overline{\mathcal{T}_3}$ and $\overline{\mathcal{T}_6}$. Hence the image is $\{1, 2pq\}$. □

Following the above proof of Lemma 3.39, one can easily obtain the next three cases.

Lemma 3.40. *If* $(p,q) \equiv (3,5) \pmod 8$, *then*

$$\overline{\alpha}\left(\overline{E_{-2pq}}(\mathbb{Q})\right) = \{1, 2pq\} \quad \textit{for both } \left(\frac{p}{q}\right) = 1 \quad \textit{and } \left(\frac{p}{q}\right) = -1.$$

Lemma 3.41. *If* $(p,q) \equiv (5,3) \pmod 8$, *then*

$$\overline{\alpha}\left(\overline{E_{-2pq}}(\mathbb{Q})\right) = \{1, 2pq\} \quad \textit{for both } \left(\frac{p}{q}\right) = 1 \quad \textit{and } \left(\frac{p}{q}\right) = -1.$$

Lemma 3.42. *If* $(p,q) \equiv (5,5) \pmod 8$, *then*

$$\overline{\alpha}\left(\overline{E_{-2pq}}(\mathbb{Q})\right) = \{1, 2pq\} \quad \textit{for both } \left(\frac{p}{q}\right) = 1 \quad \textit{and } \left(\frac{p}{q}\right) = -1.$$

4. Proof of Theorems 1.1-1.4

Now using §2 and §3, we prove Theorems 1.1-1.4.

Proof of Theorem 1.1. Let $r = \operatorname{rank}(E_{-2pq}(\mathbb{Q}))$. By Lemmas 3.6 and 3.27,

$$|\alpha(E_{-2pq}(\mathbb{Q}))| \le 16, \quad \left|\overline{\alpha}\left(\overline{E_{-2pq}}(\mathbb{Q})\right)\right| \le 8, \quad \text{for } \left(\frac{p}{q}\right) = 1,$$

$$|\alpha(E_{-2pq}(\mathbb{Q}))| \le 8, \quad \left|\overline{\alpha}\left(\overline{E_{-2pq}}(\mathbb{Q})\right)\right| \le 4, \quad \text{for } \left(\frac{p}{q}\right) = -1.$$

Therefore, Proposition 2.1 gives $r \le 5$ for $\left(\frac{p}{q}\right) = 1$, and $r \le 3$ for $\left(\frac{p}{q}\right) = -1$. By Lemma 2.1, $w(E_{-2pq}/\mathbb{Q}) = -1$. Hence, assuming the Parity Conjecture, r is odd. Consequently,

$$\operatorname{rank}(E_{-2pq}(\mathbb{Q})) = \begin{cases} 1, 3, \text{ or } 5, & \text{for } \left(\frac{p}{q}\right) = 1, \\ 1 \text{ or } 3, & \text{for } \left(\frac{p}{q}\right) = -1. \end{cases}$$

Proof of Theorem 1.2.

Proof. For the six residue classes occurring in the theorem, the relevant pairs of lemmas are

$(p,q) \pmod 8$	lemmas used
$(1,3)$	3.10, 3.31
$(3,1)$	3.14, 3.35
$(1,5)$	3.11, 3.32
$(5,1)$	3.16, 3.37
$(1,7)$	3.7, 3.28
$(7,1)$	3.8, 3.29

In every row, the corresponding lemmas give

$$|\alpha(E_{-2pq}(\mathbb{Q}))| \le 8, \quad \left|\overline{\alpha}\left(\overline{E_{-2pq}}(\mathbb{Q})\right)\right| \le 4, \quad \text{for } \left(\frac{p}{q}\right) = 1,$$

$$|\alpha(E_{-2pq}(\mathbb{Q}))| \le 4, \quad \left|\overline{\alpha}\left(\overline{E_{-2pq}}(\mathbb{Q})\right)\right| = 2, \quad \text{for } \left(\frac{p}{q}\right) = -1.$$

Thus Proposition 2.1 yields $r \le 3$ for $\left(\frac{p}{q}\right) = 1$, $r \le 1$ for $\left(\frac{p}{q}\right) = -1$. By Lemma 2.1 and the Parity Conjecture, r is odd. Hence

$$\operatorname{rank}(E_{-2pq}(\mathbb{Q})) = \begin{cases} 1 \text{ or } 3, & \text{for } \left(\frac{p}{q}\right) = 1, \\ 1, & \text{for } \left(\frac{p}{q}\right) = -1. \end{cases}$$

□

Proof of Theorem 1.3.

Proof. For the eight residue classes in the theorem, use the following pairs of lemmas:

$(p, q) \pmod 8$	lemmas used
$(3, 3)$	3.18, 3.39
$(3, 5)$	3.19, 3.40
$(5, 3)$	3.20, 3.41
$(5, 5)$	3.21, 3.42
$(3, 7)$	3.15, 3.36
$(7, 3)$	3.12, 3.33
$(5, 7)$	3.17, 3.38
$(7, 5)$	3.13, 3.34

For either value of $\left(\frac{p}{q}\right)$, the cited lemmas give $|\alpha(E_{-2pq}(\mathbb{Q}))| \le 4$, and $\left|\overline{\alpha}\left(\overline{E_{-2pq}}(\mathbb{Q})\right)\right| = 2$. Hence Proposition 2.1 gives $r \le 1$. On the other hand, Lemma 2.1 and the Parity Conjecture imply that r is odd. Therefore $r = 1$ in all eight residue classes, for both $\left(\frac{p}{q}\right) = 1$ and $\left(\frac{p}{q}\right) = -1$. □

Proof of Theorem 1.4.

Proof. Assume $(p, q) \equiv (7, 7) \pmod 8$. By Lemmas 3.9 and 3.30, we obtain for both $\left(\frac{p}{q}\right) = 1$ and $\left(\frac{p}{q}\right) = -1$, $|\alpha(E_{-2pq}(\mathbb{Q}))| \le 8$, and $\left|\overline{\alpha}\left(\overline{E_{-2pq}}(\mathbb{Q})\right)\right| \le 4$. Therefore Proposition 2.1 gives $r \le 3$. Since $w(E_{-2pq}/\mathbb{Q}) = -1$ by Lemma 2.1, the Parity Conjecture implies that r is odd. Thus $\operatorname{rank}(E_{-2pq}(\mathbb{Q})) = 1$ or 3.

□

5. Proof of Theorem 1.5

In each case, we exhibit an integral solution of the relevant torsor. The corresponding lemmas determine the images of α and $\overline{\alpha}$, and Proposition 2.1 then gives $\operatorname{rank}(E_{-2pq}(\mathbb{Q})) = 1$.
(i) For $(p, q) \equiv (3, 3) \pmod 8$, substituting $M = 2$ and $e = 1$ into $\mathcal{T}_3$ gives

$$N^2 = -2M^4 + pqe^4 = pq - 32.$$

Thus, $\mathcal{T}_3$ has the integral solution $(N, M, e) = (\sqrt{pq - 32}, 2, 1)$. By Lemmas 3.18 and 3.39,

$$\alpha(E_{-2pq}(\mathbb{Q})) = \{1, pq, -2, -2pq\}, \qquad \overline{\alpha}(\overline{E_{-2pq}}(\mathbb{Q})) = \{1, 2pq\}.$$

Hence, the cardinalities of these two sets are 4 and 2, respectively. Proposition 2.1 gives the result for both $\left(\frac{p}{q}\right) = 1$ and $\left(\frac{p}{q}\right) = -1$.

p	q	$N = \sqrt{pq - 32}$	$\operatorname{rank}(E_{-2pq}(\mathbb{Q}))$
11	3	1	1
19	3	5	1
43	11	21	1

TABLE 4. Examples corresponding to part (i).

(ii) Suppose that $(p, q) \equiv (3, 5) \pmod 8$. For $\left(\frac{p}{q}\right) = 1$, putting $M = e = 1$ in $\mathcal{T}_6$ gives $N^2 = p - 2q$; hence $(N, M, e) = (\sqrt{p - 2q}, 1, 1)$ is an integral solution. For $\left(\frac{p}{q}\right) = -1$, putting $M = e = 1$ in $\mathcal{T}_5$ gives $N^2 = 2p - q$; hence $(N, M, e) = (\sqrt{2p - q}, 1, 1)$ is an integral solution. The result now follows from Lemmas 3.19 and 3.40, together with Proposition 2.1.

$\left(\frac{p}{q}\right) = 1$				$\left(\frac{p}{q}\right) = -1$			
p	q	$N = \sqrt{p - 2q}$	Rank	p	q	$N = \sqrt{2p - q}$	Rank
11	5	1	1	11	13	3	1
19	5	3	1	19	13	5	1
59	5	7	1	43	5	9	1

TABLE 5. Examples corresponding to part (ii).

(iii) Suppose that $(p, q) \equiv (5, 3) \pmod 8$. For $\left(\frac{p}{q}\right) = 1$, putting $M = e = 1$ in $\mathcal{T}_4$ gives $N^2 = q - 2p$, while for $\left(\frac{p}{q}\right) = -1$, putting $M = e = 1$ in $\mathcal{T}_7$ gives $N^2 = 2q - p$. Thus, the relevant integral solutions are

$$(\sqrt{q - 2p}, 1, 1) \quad \text{and} \quad (\sqrt{2q - p}, 1, 1),$$

respectively. Equivalently, both conclusions follow from part (ii) by interchanging p and q.

$\left(\frac{p}{q}\right) = 1$				$\left(\frac{p}{q}\right) = -1$			
p	q	$N = \sqrt{q - 2p}$	Rank	p	q	$N = \sqrt{2q - p}$	Rank
5	11	1	1	13	11	3	1
5	19	3	1	13	19	5	1
5	59	7	1	5	43	9	1

TABLE 6. Examples corresponding to part (iii).

(iv) Suppose that $(p, q) \equiv (3, 7) \pmod 8$. For $\left(\frac{p}{q}\right) = 1$, putting $M = 1$ and $e = 2$ in $\mathcal{T}_5$ gives

$$N^2 = -qM^4 + 2pe^4 = 32p - q,$$

and hence the integral solution $(N, M, e) = (\sqrt{32p - q}, 1, 2)$. For $\left(\frac{p}{q}\right) = -1$, putting $M = e = 1$ in $\mathcal{T}_4$ gives $N^2 = q - 2p$ and the integral solution $(N, M, e) = (\sqrt{q - 2p}, 1, 1)$. The corresponding descent lemmas and Proposition 2.1 give the result.

(v) Suppose that $(p, q) \equiv (7, 3) \pmod 8$. For $\left(\frac{p}{q}\right) = 1$, putting $M = e = 1$ in $\mathcal{T}_6$ gives $N^2 = p - 2q$ and the solution $(N, M, e) = (\sqrt{p - 2q}, 1, 1)$. For $\left(\frac{p}{q}\right) = -1$, putting $M = 1$ and

$\left(\frac{p}{q}\right)=1$				$\left(\frac{p}{q}\right)=-1$			
p	q	$N=\sqrt{32p-q}$	Rank	p	q	$N=\sqrt{q-2p}$	Rank
3	71	5	1	3	7	1	1
11	271	9	1	11	23	1	1
19	599	3	1	11	71	7	1

TABLE 7. Examples corresponding to part (iv).

$e=2$ in $\mathcal{T}_7$ gives $N^2=32q-p$ and the solution $(N,M,e)=(\sqrt{32q-p},1,2)$. Equivalently, this follows from part (iv) by interchanging p and q.

$\left(\frac{p}{q}\right)=1$				$\left(\frac{p}{q}\right)=-1$			
p	q	$N=\sqrt{p-2q}$	Rank	q	p	$N=\sqrt{32q-p}$	Rank
7	3	1	1	3	71	5	1
23	11	1	1	11	271	9	1
71	11	7	1	19	599	3	1

TABLE 8. Examples corresponding to part (v).

(vi) Suppose that $(p,q)\equiv(5,7)\pmod 8$. For $\left(\frac{p}{q}\right)=1$, putting $M=1$ and $e=2$ in $\mathcal{T}_5$ gives $N^2=32p-q$ and the solution $(N,M,e)=(\sqrt{32p-q},1,2)$. For $\left(\frac{p}{q}\right)=-1$, putting $M=e=1$ in $\mathcal{T}_7$ gives $N^2=2q-p$ and the solution $(N,M,e)=(\sqrt{2q-p},1,1)$. The corresponding descent lemmas and Proposition 2.1 give the result.

$\left(\frac{p}{q}\right)=1$				$\left(\frac{p}{q}\right)=-1$			
p	q	$N=\sqrt{32p-q}$	Rank	p	q	$N=\sqrt{2q-p}$	Rank
5	79	9	1	5	7	3	1
13	191	15	1	13	31	7	1
13	367	7	1	61	71	9	1

TABLE 9. Examples corresponding to part (vi).

(vii) Suppose that $(p,q)\equiv(7,5)\pmod 8$. For $\left(\frac{p}{q}\right)=1$, putting $M=e=1$ in $\mathcal{T}_5$ gives $N^2=2p-q$ and the solution $(N,M,e)=(\sqrt{2p-q},1,1)$. For $\left(\frac{p}{q}\right)=-1$, putting $M=1$ and $e=2$ in $\mathcal{T}_7$ gives $N^2=32q-p$ and the solution $(N,M,e)=(\sqrt{32q-p},1,2)$. Equivalently, the result follows from part (vi) by interchanging p and q.

$\left(\frac{p}{q}\right)=1$				$\left(\frac{p}{q}\right)=-1$			
p	q	$N=\sqrt{2p-q}$	Rank	p	q	$N=\sqrt{32q-p}$	Rank
79	5	9	1	7	5	3	1
191	13	15	1	31	13	7	1
367	13	7	1	71	61	9	1

TABLE 10. Examples corresponding to part (vii).

(viii) Suppose that $(p,q)\equiv(5,5)\pmod 8$ and that $2pq-1$ is a perfect square. Putting $M=e=1$ in $\mathcal{T}_1$ gives

$$N^2=-M^4+2pqe^4=2pq-1.$$

Thus, $\mathcal{T}_1$ has the integral solution $(N,M,e)=(\sqrt{2pq-1},1,1)$. The corresponding descent lemmas and Proposition 2.1 give the result for both $\left(\frac{p}{q}\right)=1$ and $\left(\frac{p}{q}\right)=-1$.

□

p	q	$N = \sqrt{2pq-1}$	$\text{rank}(E_{-2pq}(\mathbb{Q}))$
5	29	17	1
13	37	31	1

TABLE 11. Examples corresponding to part (viii).

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